Unit -5

Working Capital Management

Introduction to Working Capital Management

In the financial management of any business, working capital plays a critical role in ensuring smooth day-to-day operations. Working capital refers to the difference between current assets and current liabilities, and it represents a firm's short-term financial health and operational efficiency. Effective management of working capital is essential because it directly impacts liquidity, profitability, and risk.

Understanding the **concept and importance of working capital** helps managers strike a balance between maintaining enough resources for operations and avoiding excessive idle funds. The actual **working capital needs** of a firm are influenced by several determinants such as business size, nature of operations, seasonality, production cycle, and market conditions.

Closely connected to this is the **operating cycle and cash conversion cycle**, which track how quickly a business converts its resources into cash inflows. These cycles highlight the efficiency of asset utilization and cash flow management. Within this framework, **inventory management techniques** ensure that stock levels are optimized to prevent both shortages and excesses, while **receivables management and credit policy** determine how credit is extended to customers and how quickly collections are made. On the other side, **payables management and trade credit** involve negotiating favorable credit terms with suppliers to support cash flow.

Since liquidity is at the heart of working capital management, effective **cash management and liquidity analysis** are necessary to maintain adequate funds for obligations without holding too much unproductive cash. Finally, businesses must also make decisions on **working capital financing and**

sources, balancing short-term and long-term funding options to minimize cost and risk.

In sum, working capital management is about managing the firm's current assets and liabilities in an integrated manner. It ensures continuous business operations, strengthens financial stability, and enhances profitability through efficient utilization of resources.

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1. Concept and Importance of Working Capital

Meaning:

Working Capital represents the **difference between current assets and current liabilities** of a business.

Working Capital = Current Assets - Current Liabilities

Current Assets:

Assets that are expected to be converted into cash within one accounting year, such as:

- Cash and cash equivalents
- Accounts receivable (debtors)
- Inventory (stock)
- Marketable securities
- Prepaid expenses

Current Liabilities:

Obligations due within a year, such as:

- Accounts payable (creditors)
  Short-term loans or overdrafts
- Outstanding expenses
- Bills payable

Example:

A company named **“XYZ Ltd.”** has the following financial data:

Particulars	Amount (₹)
Cash	50,000
Accounts Receivable	1,20,000
Inventory	1,80,000
Accounts Payable	1,00,000
Outstanding Expenses	50,000

$$\text{Working Capital} = (50,000 + 1,20,000 + 1,80,000) - (1,00,000 + 50,000)$$

$$\text{Working Capital} = 3,50,000 - 1,50,000 = ₹2,00,000$$

So, **Net Working Capital = ₹2,00,000**, meaning XYZ Ltd. has enough short-term assets to pay its short-term obligations.

Types of Working Capital:

1. **Gross Working Capital:**
   Total value of current assets.
2. **Net Working Capital:**
   The excess of current assets over current liabilities (CA – CL).

- Positive NWC → Healthy liquidity.
- Negative NWC → Liquidity issues or over-dependence on short-term borrowing.

Importance of Working Capital in Corporate Finance

Working capital management ensures a company's short-term financial stability and operational efficiency. It affects **liquidity, profitability, and risk**.

(i) Ensures Liquidity

A firm with adequate working capital can easily meet its day-to-day obligations such as paying wages, suppliers, and short-term loans — ensuring smooth business operations.

(ii) Smooth Business Operations

Availability of working capital allows continuous production and sales without interruption due to lack of materials or cash.

(iii) Improves Profitability

Efficient working capital management reduces financing costs, avoids stock-outs, and enhances sales — leading to better profitability.

(iv) Strengthens Solvency and Creditworthiness

Companies with strong working capital are viewed favorably by banks, creditors, and investors as financially stable and reliable.

(v) Facilitates Expansion and Growth

Adequate working capital enables firms to seize market opportunities and expand operations without liquidity strain.

(vi) Reduces Business Risk

Proper balance between liquidity and profitability reduces insolvency risk and ensures long-term survival.

Importance	Explanation	Example
Liquidity	Helps meet short-term obligations easily.	XYZ Ltd. pays suppliers on time and gets discounts for early payments.
Smooth Operations	Ensures production and sales without interruptions.	A car manufacturer can keep producing cars because it always has money to buy steel and spare parts.
Profitability	Efficient management reduces idle funds.	A retailer maintains optimal stock, avoiding excess inventory that locks up cash.
Solvency & Creditworthiness	Firms with adequate working capital can borrow easily.	A company with good liquidity easily gets a bank loan for expansion.
Growth	Supports expansion and investment in new projects.	A textile firm uses surplus working capital to open a new showroom.

Risk Reduction	Prevents financial distress and business failure.	A construction firm with enough liquidity can pay labor and material suppliers even if payments from clients are delayed.

Determinants of Working Capital Needs

Working capital requirements vary depending on several **internal** and **external** factors.

A. Nature of Business

- **Manufacturing firms** require large working capital (for raw materials, WIP, finished goods).
- **Trading firms** need moderate working capital (focus on inventory & receivables).
- **Service firms** require less working capital (no large inventory).
- **Example:** A **steel manufacturing firm** needs to buy raw materials (iron ore, coal) and maintain large inventories → high working capital.
- A **consulting firm** (service-based) has minimal inventory → low working capital.

B. Business Cycle

During **boom periods**, higher production and sales increase working capital needs.
During **recession**, sales drop and working capital requirement decreases.

- During an **economic boom**, a company like **Maruti Suzuki** increases production → needs more cash for materials, labor, etc.

- In a **recession**, production and sales slow down → less working capital needed.

C. Production Cycle / Operating Cycle

The time gap between purchase of raw material and realization of sales in cash.
Longer the cycle → higher working capital required.
Shorter cycle → lower requirement.

Operating Cycle = Raw Material Period + WIP Period+ Finished Goods Period + Debtors Period - Creditors Period

- **Example:** A **shipbuilding company** has a long production cycle (months or years) → higher working capital.

- A **bakery** produces and sells daily → short cycle → lower working capital.

D. Credit Policy

- **Liberal credit policy** (more credit sales) → higher receivables → higher working capital.

- **Strict credit policy** → quick cash inflows → lower working capital.
- **Example:** If **Samsung** offers customers 60 days' credit to boost sales, it must finance receivables → higher working capital.

- If it switches to cash-only sales, the requirement drops

E. Growth and Expansion

As the business grows, investment in inventory, receivables, and other current assets rises — requiring more working capital.

Example: Reliance Retail expanding into new cities needs more inventory, more staff, and more receivables → higher working capital.

F. Seasonal Variations

Businesses with seasonal demand (e.g., textile, sugar, or AC manufacturing) need additional working capital during peak seasons.

- **Example: Air conditioner manufacturers** need more working capital in summer season due to higher production and sales.

- **Umbrella makers** need more before monsoon.

G. Availability of Raw Materials

- **Easy availability** reduces the need for large inventories.

- **Scarce or irregular supply** requires maintaining large stocks → higher working capital.
  Example: A **sugar factory** buys sugarcane only during the harvest season, so it must store large quantities — increasing working capital.

- A **bottled water company** gets raw materials year-round, so less capital is needed.

H. Level of Technology and Automation

Automation shortens the production cycle and reduces inventory holding, lowering working capital requirements.

- **Example: Automated plants** like Toyota use just-in-time (JIT) production → lower inventory → less working capital.

- Traditional plants maintain higher stock → more working capital.

I. Profitability and Retained Earnings

Firms earning high profits generate internal cash and depend less on external working capital financing.

Example: Infosys generates strong cash flows from operations, so it can finance working capital internally.
A loss-making startup must borrow more → higher external working capital need.

J. Operating Efficiency

Efficient inventory management, cost control, and timely collection of receivables help minimize working capital needs.

Example: If **Big Bazaar** improves stock management, sells goods faster, and collects payments quickly, its working capital need reduces.

K. Inflation

Inflation raises prices of raw materials and other inputs, increasing the working capital required for the same volume of operations.

Example: If raw material prices rise (e.g., steel or fuel), a construction company needs more funds to buy the same quantity — higher working capital requirement.

L. Market Conditions

Competitive markets often lead to extended credit periods and higher inventory levels, increasing working capital needs.

- **Example:** In a highly competitive mobile phone market, firms offer longer credit periods to distributors → receivables
- increase → higher working capital.

3. Operating Cycle and Cash Conversion Cycle

1. Operating Cycle (OC)

Meaning:

The **Operating Cycle** refers to the total time taken by a business to convert its **inventory and other inputs into cash**.
It represents the period between the purchase of raw materials and the collection of cash from sales.

In simple words:
🕒 **Operating Cycle = Time from buying raw materials → producing goods → selling → receiving cash.**

Components of Operating Cycle:

1. **Inventory Conversion Period (ICP)**
    - Time taken to convert raw materials into finished goods and sell them.
    - Formula:

$$ICP = \frac{Average Inventory}{Cost of Goods Sold} \times 365$$

2. **Receivables Conversion Period (RCP)**
    - Time taken to collect cash from customers after sale.

- Formula:

$$RCP = \frac{Average Accounts Receivable}{Credit Sales} \times 365$$

3. **Payables Deferral Period (PDP)**
   - Time taken to pay suppliers after purchasing raw materials.
   - Formula:

$$PDP = \frac{Average Accounts Payable}{Credit Purchases} \times 365$$

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Formula for Operating Cycle:

Operating Cycle=ICP+RCP−PDP

Example:

Particulars	Days
Inventory Conversion Period	60
Receivables Conversion Period	30
Payables Deferral Period	40

OC=60+30−40=50 days

So, it takes **50 days** to convert investment in inventory into cash.

Importance of Operating Cycle:

- Helps in **determining working capital needs**.
- Shorter cycle → **faster cash recovery** → better liquidity.
- Longer cycle → **more capital tied up** → higher financing cost.

2. Cash Conversion Cycle (CCC)

Meaning:

The **Cash Conversion Cycle** (also called **Net Operating Cycle**) measures how long it takes for a firm to convert its **cash outflows** into **cash inflows** through operations.

It shows the number of days between **cash paid to suppliers** and **cash received from customers**.

Formula:

CCC=Inventory Conversion Period+Receivables Conversion Period−Payables Deferral Period

Note: This is numerically same as Operating Cycle but conceptually focuses only on **cash movement**.

Example:

Using previous data:

CCC=60+30−40=50 days

It means the firm's cash is tied up for 50 days in the production and sales process before it is recovered.

Significance:

- Lower CCC → better liquidity and cash flow management.
- Helps managers **identify bottlenecks** in inventory, receivables, or payables.
- Short CCC → reduces need for short-term financing.

Inventory Management Techniques

Effective inventory management ensures that there is **no overstocking or stockouts**, balancing cost and availability.

(a) Economic Order Quantity (EOQ):

- Determines the **optimal quantity** of inventory to order at one time.

- Minimizes **total inventory cost** (ordering + holding cost).
  Formula:

  Formula:

  $$EOQ = \sqrt{\frac{2DS}{H}}$$

  where:

  D = Annual demand (units)

  S = Ordering cost per order

  H = Holding cost per unit per year

  Example:

  If annual demand = 10,000 units, ordering cost = ₹100/order, holding cost = ₹2/unit

  $$EOQ = \sqrt{\frac{2 \times 10,000 \times 100}{2}} = 1000 \text{ units}$$

(b) ABC Analysis:

- Categorizes inventory based on value and importance:
  - **A items:** High value, low quantity → strict control.
  - **B items:** Moderate value and control.
  - **C items:** Low value, high quantity → simple control.
- Used for **prioritizing management attention**.

(c) Just-In-Time (JIT):

- Inventory is ordered **only when needed** for production.
- Reduces carrying cost, but requires **strong supplier coordination**.
- Common in **Toyota Production System**.

(d) FIFO, LIFO, and Weighted Average:

- **FIFO (First-In, First-Out):** Older stock sold first; reflects realistic flow.
- **LIFO (Last-In, First-Out):** Latest stock sold first; used during inflation to reduce tax liability.
- **Weighted Average:** Average cost method; smoothens price fluctuations.

(e) Reorder Point (ROP):

- Level of inventory when new order must be placed to avoid stockouts. ROP=Lead Time Demand+Safety Stock

(f) Safety Stock:

- Extra stock kept to prevent shortages due to demand or supply dela

(g) Inventory Turnover Ratio:

Indicates how efficiently inventory is managed.

$$\text{Inventory Turnover Ratio} = \frac{Cost of Goods Sold}{Average Inventory}$$

Higher ratio → faster movement → efficient management.

Summary Table

Concept	Meaning	Formula	Focus
Operating Cycle	Time to convert resources into cash	ICP + RCP - PDP	Overall process
Cash Conversion Cycle	Time cash remains tied up	ICP + RCP - PDP	Cash perspective
EOQ	Optimum order size	√(2DS/H)	Cost minimization
ABC Analysis	Inventory classification	A-B-C	Control level
JIT	Order as needed	—	Zero inventory
ROP	Reorder level	Lead time demand + safety stock	Stock availability

Receivables Management and Credit Policy

A. Receivables Management

Definition:
Receivables are amounts due from customers for credit sales. Managing

them efficiently ensures good customer relations while maintaining liquidity and profitability.

Objectives:

1. To increase sales by extending credit to reliable customers.
2. To minimize bad debts and overdue accounts.
3. To optimize the investment in receivables.

Importance:

- Too liberal a policy → increases sales but may cause **liquidity problems**.
- Too strict a policy → reduces bad debts but may **lose customers**.

Example:

Suppose **Tata Motors** sells commercial vehicles to fleet operators on a 60-day credit basis.

- If Tata offers longer credit (90 days), sales may rise.
- But receivables also rise, increasing the need for working capital.
- Management must analyze whether **additional profit** from increased sales justifies the **extra cost of carrying receivables**.

Key Ratios:

1. **Average Collection Period (ACP):**
   [
   \text{ACP} = \frac{\text{Average Accounts Receivable}}{\text{Credit

Sales per Day}}
]
→ Shorter ACP = faster collection.

2. **Receivables Turnover Ratio:**
   [
   \text{Net Credit Sales} / \text{Average Receivables}
   ]
   → Higher ratio = efficient collection.

B. Credit Policy

Purpose:
A good credit policy maintains a balance between **risk** and **returns**.

Components:

1. **Credit Standards** – Criteria for approving customers.
    - Example: Credit score, past payment history.
2. **Credit Terms** – Conditions like credit period and cash discounts.
    - Example: *2/10, net 30* → 2% discount if paid in 10 days, full payment due in 30.
3. **Collection Policy** – Steps for overdue payments (reminders, interest, legal action).
4. **Credit Limit** – Maximum credit allowed per customer.

Example:

Samsung Electronics might allow its distributors 45 days' credit, but if a dealer frequently delays payments, the credit limit may be reduced to 30 days or cash terms.

Payables Management and Trade Credit

A. Payables Management

Definition:
It is the process of managing obligations towards suppliers (creditors). Proper management helps maintain supplier relations and ensure liquidity.

Objectives:

- To make payments within due dates.
- To optimize the use of available credit.
- To maintain a balance between cash availability and supplier trust.

Example:

If **Maruti Suzuki** buys auto parts worth ₹10 crore from a supplier with 60 days' credit:

- Paying early (within 30 days) might give a 2% discount (₹20 lakh).
- Delaying payment beyond 60 days may attract penalty or spoil relations.
  Hence, they'll analyze the **cost-benefit** before choosing when to pay.

Key Ratios:

1. **Payables Turnover Ratio:**
   [
   \text{Net Credit Purchases} / \text{Average Accounts Payable}
   ]
2. **Average Payment Period (APP):**
   [

365 / \text{Payables Turnover}
]

B. Trade Credit

Definition:
It's a short-term financing option provided by suppliers. It's a **spontaneous source** of working capital.

Forms of Trade Credit:

1. **Open Account:** Supplier sends goods and bills later.
2. **Bills Payable:** Buyer signs a bill promising future payment.
3. **Promissory Note:** A written promise to pay at a later date.

Example:

A textile company **Raymond Ltd.** purchases fabric from a mill on "net 60" terms — payment due in 60 days.
This allows Raymond to **use and sell fabric first** and **pay later**, easing cash flow.

Advantages:

- Easy and interest-free (if paid on time).
- Builds supplier relationships.
- Flexible with business volume.

Disadvantages:

- Discounts lost if not paid early.
- May affect credit reputation if delayed.

Cash Management and Liquidity Analysis

A. Cash Management

Definition:
Cash management involves planning and controlling cash inflows and outflows to maintain adequate liquidity.

Objectives:

- Ensure availability of cash when needed.
- Reduce idle funds (earning nothing).
- Optimize cash between operations, investments, and financing.

Motives for Holding Cash (Keynesian View):

1. **Transaction Motive:** To pay for routine expenses.
2. **Precautionary Motive:** For emergencies or unexpected needs.
3. **Speculative Motive:** To take advantage of opportunities (e.g., buy shares at low price).

Cash Management Techniques:

1. **Cash Budgeting:** Forecasting future cash needs.

2. **Synchronizing Cash Flows:** Matching inflows and outflows.

3. **Float Management:** Reducing the time between cheque issue and clearance.

4. **Lock Box System:** Customers deposit payments directly in company's bank.

5. **Investment of Surplus Cash:** In short-term marketable securities (e.g., Treasury bills).

Example:

Infosys Ltd. maintains large cash reserves for payroll and contingencies. It invests surplus cash in liquid mutual funds and Treasury bills to earn short-term returns without affecting liquidity.

B. Liquidity Analysis

Definition:
Liquidity indicates a firm's ability to meet its short-term obligations as they become due.

Ratios Used:

Ratio	Formula	Interpretation
Current Ratio	Current Assets / Current Liabilities	Measures overall liquidity (Ideal = 2:1)
Quick Ratio	(Current Assets − Inventory) / Current Liabilities	Excludes slow-moving inventory (Ideal = 1:1)
Cash Ratio	(Cash + Marketable Securities) / Current Liabilities	Strictest test of liquidity

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Example:

If **Hindustan Unilever Ltd.** has

- Current Assets = ₹10,00,000
- Current Liabilities = ₹5,00,000

Then,
$$
\text{Current Ratio} = 10,00,000 / 5,00,000 = 2:1
$$
→ Excellent liquidity position.

Working Capital Financing and Sources

A. Working Capital Financing

Definition:
It refers to obtaining funds to finance the day-to-day operations of a business (inventory, receivables, wages, etc.).

Types:

Basis	Type	Description
Duration	**Short-Term**	For temporary needs (less than 1 year)
	Long-Term	For permanent working capital (fixed level of current assets)

Sources of Working Capital Financing

1. Short-Term Sources:

Source	Description	Example
Trade Credit	Credit from suppliers	Buy raw materials now, pay later
Bank Overdraft / Cash Credit	Withdraw funds up to a limit	SBI overdraft facility
Commercial Paper (CP)	Unsecured promissory note issued by large companies	Reliance Industries issues CPs
Factoring	Selling receivables to a factor for cash	Bajaj Finance factoring services
Bills Discounting	Selling bills before maturity at a discount	Exporters discount bills with banks

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2. Long-Term Sources:

Source	Description	Example
Share Capital	Owners' investment	IPO / Equity issue
Debentures / Bonds	Borrowed capital	Corporate bond issue
Retained Earnings	Profits reinvested in business	Infosys internal funding
Term Loans	Long-term bank loans	For working capital margin
Venture Capital / Private Equity	Investment by financial institutions	Startups funding working capital needs

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